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Aging rice straw reduces the bioavailability of mercury and methylmercury in paddy soil

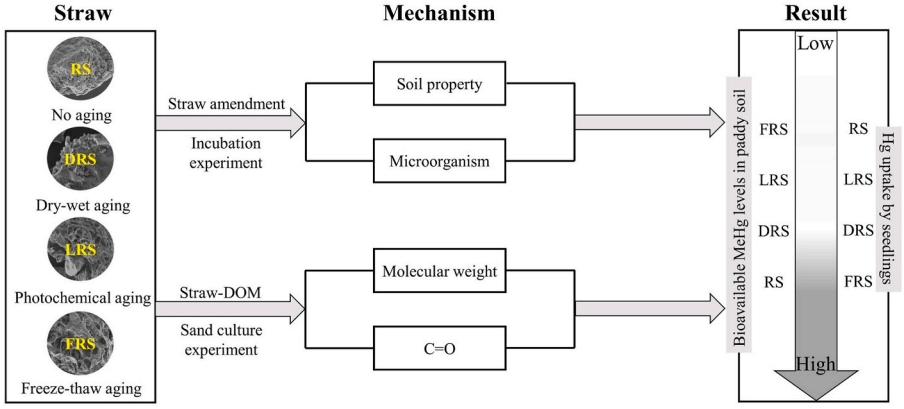
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HIGHLIGHTS

- Aging straw amendment decreased bioavailable MeHg levels in paddy soil.
- Aging straw-DOM increased bioavailability of inorganic Hg.
- Differently aging treatments of straw had varying effects on Hg methylation in paddy soil.
- The ability of freeze-thaw aged straw to inhibit Hg methylation in paddy soil was the strongest.

GRAPHICAL ABSTRACT



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ABSTRACT

Straw amendment is a prevalent agricultural practice worldwide, which can reduce air pollution and improve soil fertility. However, the impact of aging straw amendment on the bioavailability of mercury (Hg) and methylmercury (MeHg) in paddy soil remains unclear. To investigate this, incubation experiments were conducted using the diffusive gradient in thin-film technique. Results showed that amendments of dry-wet aging (DRS), photochemical aging (LRS), and freeze-thaw aging rice straw (FRS) reduced the bioavailable MeHg in paddy soil by 2.2–27.6%, 13.5–69.8%, and 23.5–86.1%, respectively, compared to fresh rice straw (RS) amendment. This result could be due to changes in soil properties such as soil pH and overlying water Fe and Mn as well as microbial abundance (including Clostridiaceae, Firmicutes, and Actinobacteriota). Simultaneously, The LRS and FRS amendments reduced bioavailable Hg in paddy soil by 20.0–40.8% and 17.1–48.6%, respectively, while DRS increased the bioavailable Hg by 15.8–120.0%. This could be attributed to changes in soil oxidation-reduction potential and overlying water SO_4^{2-} content. Additionally, the results of sand culture experiments showed that the concentrations of Hg uptake by rice seedlings were 97.1–118.2%, 28.1–35.6%, and

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198.0–217.1% higher in dissolved organic matter (DOM) derived from DRS, LRS, and FRS than RS, indicating that aging straw leached DOM may promote the Hg bioavailable when straw amendment. This result could be due to lower molecular weight and higher C═O functional group content. These results provide new insight into how aging straw amendment affects the bioavailability of Hg and MeHg in paddy soil under different climates.

1. Introduction

Mercury (Hg) pollution has attracted much attention because it can convert into the potent neurotoxin, methylmercury (MeHg), which can be biomagnified in aquatic food webs thus pose a threat to fish-eating populations (Wu et al., 2020). However, The primary MeHg exposure pathway for Chinese inland populations is the consumption of rice rather than fish (Li et al., 2015; Zhang et al., 2010). The anaerobic environment of paddy soil may play a significant role in the formation of MeHg, which leads to its accumulation in rice (Kodamatani et al., 2020). It is therefore imperative to develop efficient strategies to lower MeHg accumulation in rice.

In the past, people used to dispose of straw by burning (He et al., 2020), but now straw amendment has replaced straw burning as the primary method of straw disposal, thus reducing the environmental pollution caused by the latter and enhancing soil fertility (Shinde et al., 2022; Liu et al., 2016). Nonetheless, it has been shown that MeHg tends to accumulate in rice grains after straw amendment, while this accumulation primarily results from in situ production in paddy soil (Wang et al., 2018, 2019, 2023; Zhu et al., 2015). Recently, the impact of straw amendment on MeHg levels in paddy soil under various treatment conditions has received lots of attention. For example, Shu et al. found that dry straw incorporation increased MeHg levels in paddy soil, while straw biochar or composed straw incorporation decreased phytoavailable MeHg levels (Shu et al., 2016). Similarly, Sun et al. confirmed that powdered straw accelerated the time when the MeHg concentrations in paddy soil reached their peak level and encouraged the methylation of Hg (Sun et al., 2022). These studies emphasize the importance of straw amendment with different forms and properties on MeHg accumulate in paddy soil. However, there is little research about the effects of different aging straw amendments (straw that has been subjected to physical, chemical, and natural biological influences in the natural environment over a long period) on Hg methylation in paddy soil.

Since the natural aging process is very slow, monitoring the effects of straw amendment application is time-consuming. Therefore, various artificial accelerated aging methods have been used to accelerate aging process by replacing natural aging, such as dry-wet cycles to simulate southern China during the rainy season (Huang et al., 2022), photodegradation cycles to simulate the strong ultraviolet irradiation in the western regions (Wu et al., 2021a), and freeze-thaw cycles to simulate the cold regions in northern China (Yang et al., 2022a). The aging process affects not only the straw itself, but also the physicochemical properties of dissolved organic matter leached from straw (straw-DOM). When straw decomposes in the soil, it releases straw-DOM, which is a non-negligible factor that affects Hg methylation in paddy soil. Previous research has revealed that the concentration (Peterson et al., 2023), aromatic and molecular weight (Zhang et al., 2022) of DOM can influence the bioavailability of Hg. In addition, DOM contains numerous functional groups that are capable of forming complexes or chelating with Hg (Graham et al., 2013; Guo et al., 2012). Furthermore, straw-DOM not only improves the ability of rice roots to absorb MeHg, but also serves as a stable carbon source for soil microorganisms, enhancing the abundance and activity of microbial methylators (Tang et al., 2019). Therefore, investigating the net effect of straw-DOM on Hg bioavailable in paddy soil is critical.

The diffusive gradient in thin-film (DGT), a non-invasive tool (Yao et al., 2016), has been identified as helpful in assessing the bioavailable Hg (Yang et al., 2022b; Ndu et al., 2018) and MeHg (Neal-Walthall et al., 2022; Wang et al., 2017) to accumulate in food webs in aquatic and soil ecosystems. Besides, previous research showed that DGT devices can predict Hg uptake plants in amended soils because they are capable of imitating the process of uptake limited by diffusion, and a significant correlation between soil Hg measured by DGT and Hg uptake by plants (Turull et al., 2019a, 2019b).

Based on the above background and concerns, this study used DGT to quantify bioavailable Hg and MeHg in paddy soil, and employed three artificial aging methods (dry-wet cycles, freeze-thaw cycles, and photochemical cycles) to accelerate the aging process of rice straw. Then, a series of incubation experiments were conducted to explore (1) changes in the bioavailability of Hg and MeHg in paddy soil after different aging rice straw amendments and (2) dynamics of soil properties and Hg methylation microorganisms. Furthermore, sand culture experiments were conducted to investigate the content of Hg uptake by rice seedlings, thus reflecting the net effect of aging straw-DOM on the bioavailability of Hg in paddy soil.

2. Materials and methods

2.1. Soil and rice straw

The Hg-contaminated paddy soil used in the incubation experiments was collected from the Wukeng (WK), in the Wanshan Hg mining area, southwestern China. The fundamental properties of paddy soil are listed in Table S1.

Rice straw was collected from a rural setting near Changsha, China, and was then washed, air-dried, ground, and sieved through a 100 μm mesh to facilitate its decomposition in situ. The total Hg (THg) and MeHg concentrations in the rice straw were 2.14 ng/g and 0.81 ng/g, respectively, indicating that the input of Hg and MeHg from the straw to soil was negligible.

2.2. Aging process and characterization of rice straw

To simulate and accelerate the natural aging process of fresh rice straw (RS), artificial aging for 90 days of RS was performed using dry-wet (DRS) cycles, photochemical (LRS) cycles, and freeze-thaw (FRS) cycles (Liu et al., 2019; Quan et al., 2020; Xu et al., 2018). For the DRS cycle, 30 g of RS were saturated with ultrapure water (>18 MΩ cm) at 100% water holding capacity for 16 h and then oven dried at 60 °C for 8 h. For LRS aging treatment, XPA II photochemical reactor (Xujiang Electromechanical Inc., China) was used, a 500 W mercury lamp (UV_{365} = 100 W/m^2) was light source. RS was added into a quartz tube filled with ultrapure water, which was then placed into the XPA II photochemical reactor for 12 h. After that, the quartz tube was wrapped by tinfoil to keep 12 h dark (Wu et al., 2021b). FRS cycle was carried out by saturating the fresh rice straw (30 g) with ultrapure water and then consistently frozen at −20 °C for 12 h followed by thawing at 25 °C for 12 h.

The surface morphologies of the straw samples were observed using a JSM-6510LV scanning electron microscope (JEOL, JPN) at a scan voltage of 20 kV. More details of characterization are shown in Text. S4.

2.3. Extraction and characterization of straw-DOM

The RS, DRS, LRS, and FRS were placed in separate conical flasks and mixed with ultrapure water at a ratio of 1/20 (w/v). The conical flasks were then capped and shaken for 24 h at 200 rpm and 25 °C before being centrifugated at 4000 g for 20 min (Wen et al., 2023). The supernatant

was extracted from each conical flask and filtered through a 0.45 μm membrane filter to obtain straw-DOM (RS_{DOM}, DRS_{DOM}, LRS_{DOM}, and FRS_{DOM}). Storing the filtered supernatant samples at −20 °C in the dark until further quantified for dissolve organic carbon (DOC) concentrations and characterization.

The concentration of DOC in the solution was determined using a TOC analyzer (Shimadzu L-series, TOC-CHP, Japan). UV–vis spectrophotometer (UV-2700, Shimaz, Japan) was used to obtain UV–vis absorption spectra. $SUVA_{254}$ is calculated as A_{254}/DOC concentration (in mg-C/L) and can be used to represent the aromaticity of DOM (Ding et al., 2020). And the E_2: E_3 and E_3: E_4 were calculated as the ratio of absorption coefficient at 245–365 nm and 300–400 nm, respectively (Ren et al., 2021; Artinger et al., 2000). Micro-Fourier Transform Infrared (FTIR) spectrometer (Shimadzu Corporation, Kyoto, Japan), excitation-emission matrix (EEM) fluorescence spectroscopy using a fluorescence spectrophotometer (F-7000, Hitachi, Japan) and x-ray photoelectron spectroscopy (XPS) (Quantum 2000, Physical Electronics Co. Ltd, USA) were used to measure the changes in surface components of the straw-DOM. Additional details are provided in Text. S5.

2.4. Incubation experiments

Incubation experiments were conducted to explore the influence of aging straw amendment on the dynamics of bioavailable Hg (DGT-Hg) and bioavailable MeHg (DGT-MeHg) in paddy soil. Briefly, 10.0 g of WK soil and 0.1 g of straw (RS, DRS, LRS, and FRS) were put in 50 mL centrifuge tubes, then poured in 30 mL deionized water. Four treatments (three replicates per treatment) were implemented in the incubation experiments. 1 mM sodium pyruvate was added to each tube to stimulate the growth of soil microbes. All sealed tubes were incubated in the dark at 25 °C. Samples were taken on day 0 (i.e., 4 h after straw amendment), as well as on days 2, 6, 10, and 20. Three tubes from each treatment were selected for measurement of soil indexes, DGT-Hg, and DGT-MeHg. The overlying water (to simulate porewater in paddy soils) (Tang et al., 2019) was filtered through a 0.45 μm membrane. At the designated incubation time, oxidation-reduction potential (ORP) and pH (by HQ30d multiparameter meters, HACH, USA) in soil were measured, each centrifuge tube was then stirred to mix and left static for 10 min to let large particles settle before being measured in anaerobic chamber. Total dissolved Fe and Mn (by ICP-MS), and sulfate concentration (by turbidimetry) (Jumina et al., 2021) in the overlying water were determined. Soil organic carbon (SOC) concentrations were measured in the same way as DOC content. The introduction and details of DGT are described in the Text. S1. Moreover, paddy soil collected at 4 h and day 20 was used to assess the impact of aging straws amendment on soil microbiome and the abundance of hgcA copies (Text. S2).

2.5. Sand culture experiments

Six germinated rice seeds were sown in a polyethylene bottle filled with 250 g of quartz sand (<1 mm, washed with 30% H_2O_2 to remove organic material on the surface). Three seeds were used for determining the uptake of Hg and the other three for quantifying tip number and root length. Three bottles were used for each treatment. 400 mg DOC/L straw-DOM was added with Hoagland solution and replenished DOM to stay hydrated to simulate the real porewater situation of straw amended soil (Tipping et al., 2003). Initially, 10 nM Hg standard solution was added to the culture solution. On days 10 and 20, the absorbed Hg concentration was determined by digesting rice seedlings and quantifying tip number and root length (winRHIZO 2013e-Expression 11000XL). The remaining seedlings were immersed in 0.8 mM cysteine solution for 2 min to remove the Hg absorbed on the surface, then washed with deionized water, dried, and ground into powder for later digestion. Details of which are described in the Text. S3.

2.6. Statistical analysis

Tukey's multiple comparison tests with one-way analysis of variance (ANOVA) ($p < 0.05$) were used for statistical analysis of differences in soil or plant parameters between treatments. Canoco (window version 5.1) software was used to perform redundancy analysis (RDA) of soil indexes with DGT-Hg and DGT-MeHg. All analyses were performed via Statistical Package for the Social Sciences (SPSS), version 16.0.

2.7. Quality assurance (QA) and quality control (QC)

To prevent cross-contamination and reagent contamination, ultra-clean benches, ultralow blanks, disposable gloves and guaranteed reagents (GR) were required in the laboratory analysis. The Hg and MeHg analysis underwent quality control using triple sample analyses, method/reagent blanks, matrix spikes, and approved reference materials. Hg loss during the manipulations, such as that caused by oven drying or filtering, was calculated and shown to be minimal (5%).

3. Results and discussion

3.1. Characterization of aging rice straw

3.1.1. SEM analysis

Fig. 1a shows SEM images of fresh and aging rice straws. Compared to the RS, the DRS experienced significant fragmentation but did not increase in pore structure. Although several pores emerged after photochemical aging, FRS exhibited larger pores and cracks compared to LRS. This indicates that freeze-thaw aging causes more damage to the straw structure. This could be attributed to the water in the straw pores freezing and forming ice crystals, which then grew inside the pores, causing the straw particles to break up and enlarge the pores (Liu et al., 2018). Compared to RS, the degree of deformation gradually increases in DRS, LRS, and FRS. This may be due to a decrease in the hardness of cellulose fibers (Boakye-Boaten et al., 2016) and degradation of lignin (Feng et al., 2019) in straw after aging treatment, resulting in increased twisting and pore structure appearance.

3.1.2. FTIR spectra

As shown in Fig. 1b, the peaks intensity between wavenumbers 1300 to 1000 cm^{-1} is ranked as DRS > FRS > LRS > RS, indicating that carboxylic and phenolic groups have been formed in straw surface during the aging process (Dong et al., 2013).

3.1.3. XPS spectra

The main elements on the surface of these straws were C, O, and N. The O: C atomic ratios (an index for oxidation state) of RS, DRS, LRS, and FRS were calculated to be 0.38, 0.55, 0.55, and 0.33, respectively, indicating that DRS and LRS had formed more O-containing functional groups than RS (Table S5). However, FRS, which is often exposed to low temperatures, did not undergo significant oxidation during the freeze-thaw process.

3.2. The dynamics of DGT-Hg and DGT-MeHg

After incorporating the fresh and aged straws, DGT-Hg and DGT-MeHg levels in the paddy soil were measured at predetermined intervals (4 h, day 2, 6, 10, and 20). During the 20-day incubation period, DGT-Hg levels in paddy soil were 20.0–40.8%, and 17.1–48.6% lower in LRS and FRS treatments than that in RS treatment, but 15.8–120.0% higher in DRS treatment than that in RS treatment (Fig. 2a). However, the larger value of SOC/DGT-Hg represents the stronger coupling between SOC and DGT-Hg, as well as better ability to fix free Hg due to the presence of more sulfhydryl groups (Fink-Mercier et al., 2022; Bian et al., 2018). And the SOC/DGT-Hg value of DRS treatment was 12.56% lower than that RS treatment, while LRS and FRS treatments exhibited

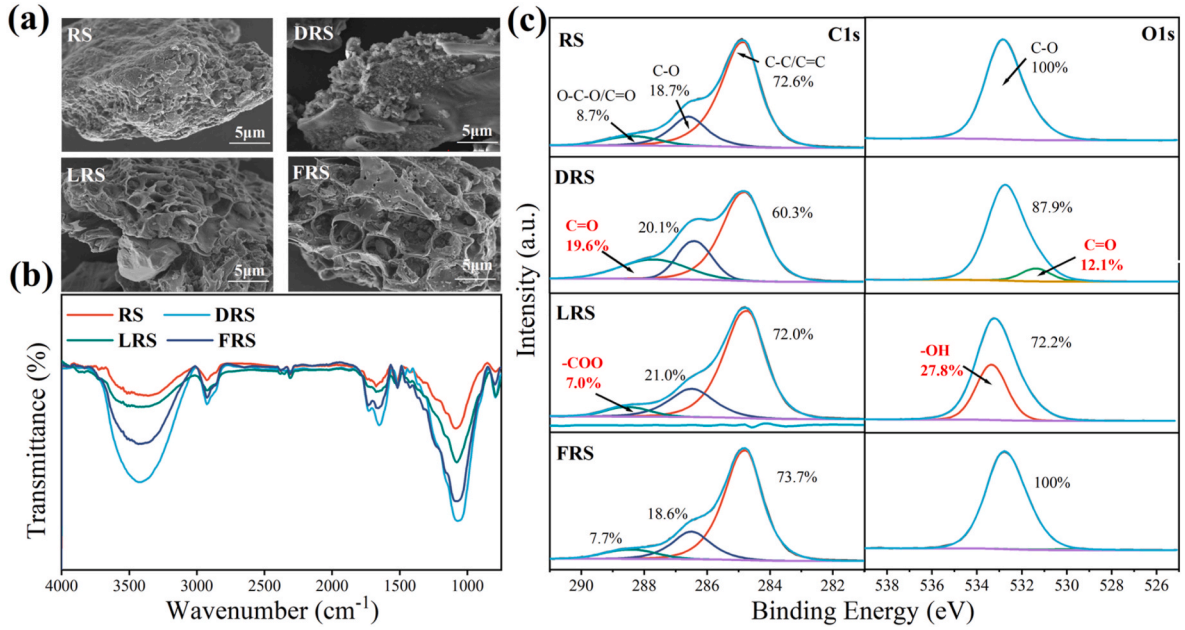


Fig. 1. Scanning electron micrographs (SEM) (a), Fourier transform infrared spectra (FTIR) (b), and X-ray photoelectron microscopy (XPS) (c) of fresh, dry-wet aging, photochemical aging, and freeze-thaw aging straw.

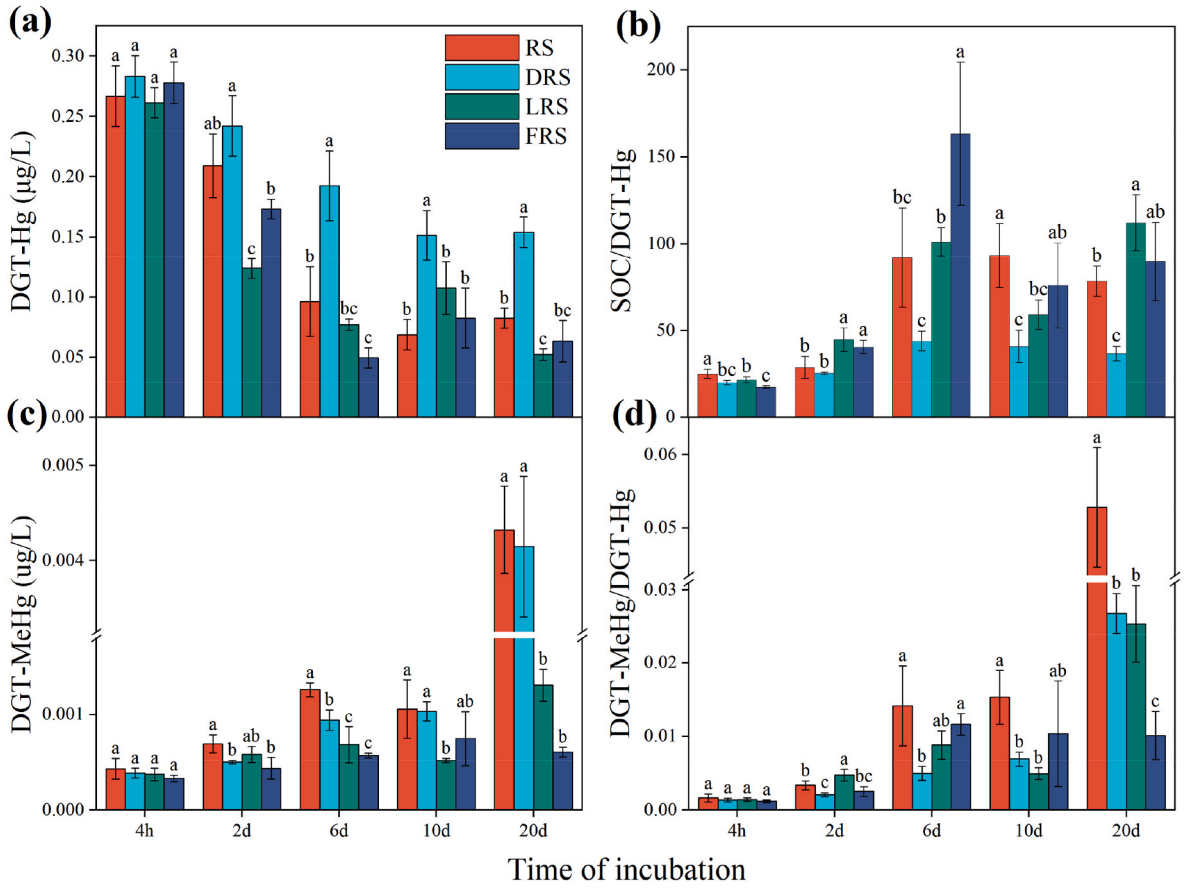


Fig. 2. Variations of DGT-Hg in paddy soil (a), the ratio of SOC to DGT-Hg (b), change of DGT-MeHg in paddy soil (c), and DGT-MeHg to Hg (d). Data are given as means ± SD (n = 3). Different letters indicate significant differences among treatments within the same day ($p < 0.05$).

an increasing trend (Fig. 2b). This could explain why LRS and FRS amendments better inhibit the bioavailability of Hg, while DRS promotes the bioavailability of Hg. As shown in Fig. 2c, the DGT-MeHg concentrations of DRS, LRS, and FRS treatments were respectively 2.2–27.6%, 13.5–69.8%, and 23.5–86.1% lower than that in RS treatment during the 20-day incubation experiments. MeHg/Hg is a common

ratio used to gauge the Hg methylation rate (Ethier et al., 2010). In the DRS, LRS, and FRS treatments, this rate was 38–65%, 38–68%, and 25–81% lower than that in RS treatment, respectively (Fig. 2d). The above results implied that artificially simulated aging straw amendment (DRS, LRS, and FRS) can effectively inhibit Hg methylation in paddy soil compared to RS amendment. The best effect was observed in the LRS and FRS treatments. Notably, after the DRS amendment, the level of DGT-Hg increased while the rate of Hg methylation decreased significantly. This phenomenon will be analyzed in section 3.3.1.

3.3. Changes in soil indexes and redundancy analysis (RDA)

To investigate the impact of aging straw on the bioavailability of Hg and MeHg in paddy soil, indicators of the soil (ORP, pH, and SOC) and overlying water (dissolved Fe and Mn, SO_4^{2-}) were measured.

3.3.1. Soil ORP

Before day 2, soil ORP of all treatment groups turned negative, indicating the formation of an anaerobic and relatively high organic soil environment due to straw decomposition consuming oxygen and releasing OM (Fig. 3a). The timing of entry into the anaerobic environment makes a difference. DRS, LRS, and FRS amendment cause the soil to enter the anaerobic environment later than RS does, which is presumably directly related to the speed of decomposition of straw. On day 20, the ORP value of DRS, LRS, and FRS treatments were 62%, 50%, and 14% higher than that of RS treatment, respectively. The soil ORP and DGT-Hg levels showed a significant positive correlation (Fig. S1a). Therefore, the increase in DGT-Hg levels after DRS amendment is perhaps mainly due to the increase in ORP values of the soil.

3.3.2. Soil pH and SOC

During incubation, the LRS treatment consistently had the highest pH, closely followed by FRS, while the RS treatment had the lowest pH (Fig. 3b). A higher soil pH indicates a more pronounced oxidizing environment, which prevents Fe and Mn from dissociating and competing with Hg for adsorption sites on soil particles. The oxidizing

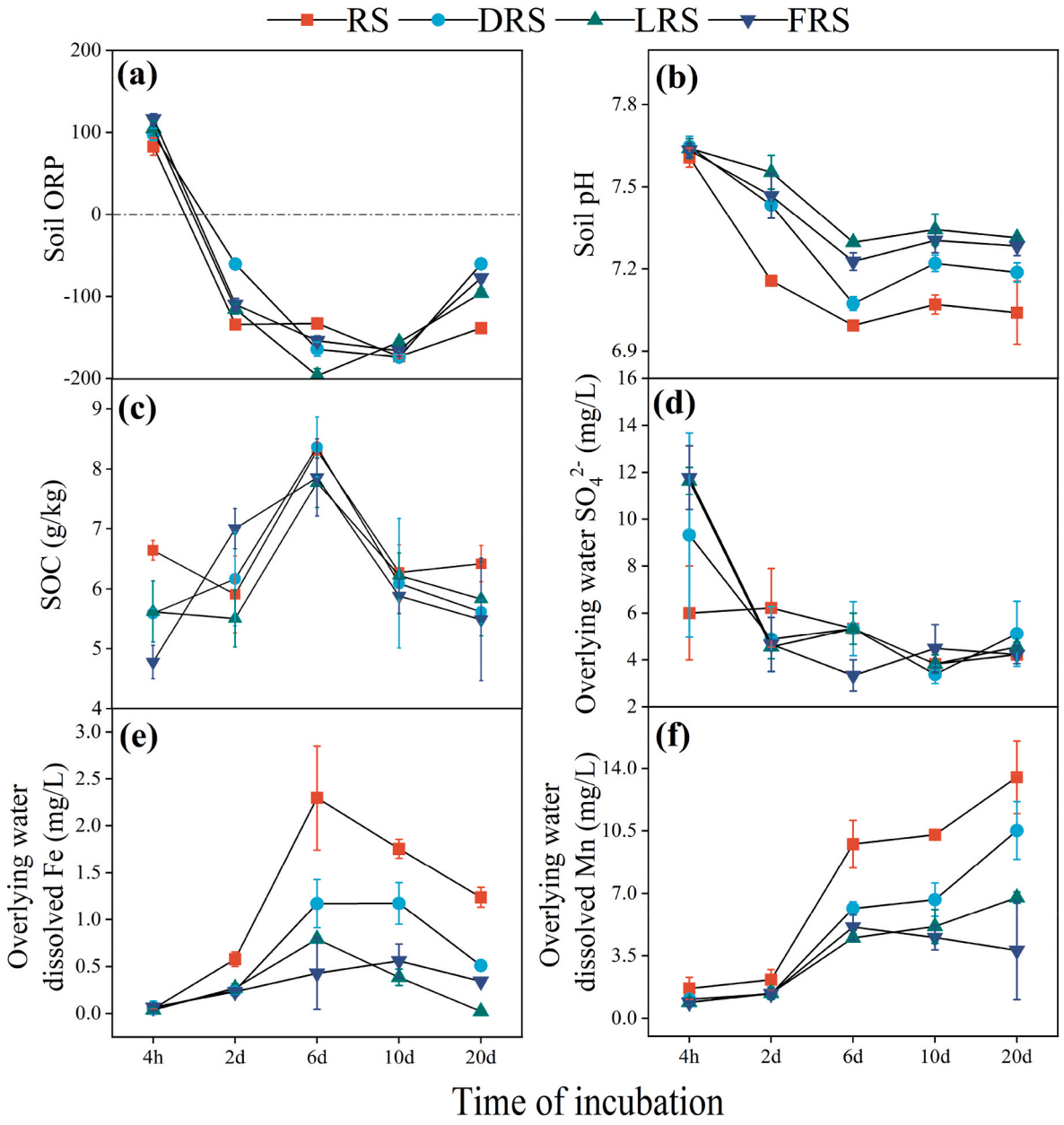


Fig. 3. Variations of ORP (a) and pH (b) in soil, SOC (c), the concentration of overlying water SO_4^{2-} (d), dissolved Fe (e), and dissolved Mn (f).

environment increased the negative charge of soil organic matter, which inclined it to form complexes with Hg, thus immobilizing Hg and decreasing its bioavailability as well as the generation of MeHg (Zhao et al., 2018; Kögel-Knabner et al., 2010). Soil pH was significantly negatively correlated with DGT-MeHg ($r = -0.57$, $p < 0.01$) (Fig. S1d). Therefore, it can be concluded that dry-wet aging, photochemical aging, and freeze-thaw aging straw amendment could influence the bioavailability of MeHg in soil by altering soil pH.

As shown in Fig. 3c, there was no significant difference in the content of SOC between RS, DRS, LRS and FRS treatments. However, they all reached their maximum concentration on day 6 simultaneously.

3.3.3. Overlying water SO_4^{2-}

The concentration of SO_4^{2-} in the DRS, LRS, and FRS treatments appeared to show a downward trend throughout the 20-day incubation period, compared to 4 h. The concentrations decreased by 29.6%, 45.2%, 60.8%, and 64.1%, respectively (Fig. 3d). According to Fig. S1b, there was a significant positive correlation between dissolved SO_4^{2-} and DGT-Hg ($r = 0.74$, $p < 0.01$). Previous reports have shown that the transportation of Hg is associated with the formation of Hg–S complexes, and S^0 can increase mobility of Hg from HgS(s) (Li et al., 2019). Therefore, our results show that the amendment of DRS, LRS, and FRS substantially reduced SO_4^{2-} in comparison to RS. This reduction may have been caused by soil microorganisms converting sulfate into S^{2-} or S^0, which could affect the bioavailability of Hg and MeHg in soil.

3.3.4. Overlying water Fe and Mn

The dissolved Fe levels (Fig. 3e) were ranked from high to low as RS, DRS, LRS, and FRS treatments during the 20-day incubation period, and the trend coincided with dissolved Mn of overlying water (Fig. 3f). The concentration of dissolved Fe peaked at day 6, except for FRS. The values of dissolved Fe for RS, DRS, LRS, and FRS treatments were 2.295, 1.171, 0.792, and 0.427 mg/L, respectively. The biggest difference was in FRS treatment, which was 81.4% lower than RS treatment, and the smallest difference was between RS and DRS treatments, which were only 49.0% lower than RS treatment. Dissolved Mn in DRS, LRS, FRS treatments decreased by 22.1–36.9%, 34.9–53.9%, and 36.1–71.9%, respectively, compared with RS treatment. Fewer dissolved Fe/Mn will compete with Hg to occupy a smaller number of binding sites (Qian et al., 2022). Consequently, both LRS and FRS amendments result in lower bioavailability of Hg compared to RS amendment. The DGT-MeHg was positively correlated with dissolved Fe ($r = 0.48$, $p < 0.05$) and significantly positively correlated with dissolved Mn ($r = 0.84$, $p < 0.01$) (Figs. S1e and f). It is concluded that DRS, LRS, and FRS amendments can all affect the soil bioavailability of MeHg by influencing dissolved Fe/Mn in the overlying water. Additionally, they may also be a key factor in reducing the soil bioavailability of Hg after LRS amendment.

3.3.5. RDA analysis

To further determine the relative importance of soil components that decide the bioavailability of Hg and MeHg in soil, redundancy analysis was undertaken (Fig. S1c). RDA1 and RDA2 contributed 59.48% and 18.99%, respectively, accounting for the top two major constituents, which together accounted for 78.47% of the variance. Overlying water Mn has the highest contribution of 68.7%, followed by soil ORP with 23.6%. This result further indicates that overlying water Mn plays a decisive role in reducing soil Hg methylation after DRS, LRS, and FRS amendment, and high soil ORP values are the key reason for the increased bioavailability of Hg after DRS amendment. To conclude, the results of soil indexes suggest that lower overlying water Fe and Mn content and higher soil pH after DRS, LRS, and FRS amendments inhibit the bioavailability of MeHg in soil. In addition, we need to further consider the impact of biological factors on the MeHg bioavailable in paddy soil after aging straw amendment.

3.4. Microbial analysis

3.4.1. Microbial community analysis

To investigate the effect of microbial factors on MeHg bioavailability variation after aging straw amendment, we performed soil microorganism sequencing. Using high-throughput sequencing analysis, we obtained 1,106,014 valid sequences with a total of 460,267,468 bases across various treatment groups. Moreover, Shannon, Chao, Simpson, and ACE indices were used to analyze the abundance and diversity of microbiome under different treatments (Table S7). Compared to the RS treatment, the Shannon index, which represents the diversity of species in the community, increased by 5.58%, 5.63%, and 5.93% for DRS, LRS, and FRS treatments, respectively. The Chao and ACE indices reflect community abundance, and the results showed that the abundance of microbial communities in DRS, LRS, and FRS treatments all increased to varying degrees. The Simpson index reflects community evenness, with RS and LRS treatments exhibiting the highest level of community evenness, while the DRS treatment shows the lowest level. These results showed that DRS, LRS, and FRS amendments increased the diversity and overall abundance of the microbial communities in soil. At the same time, DRS and FRS treatments decreased the community evenness, indicating that all aging straw amendment treatment groups could stimulate microbial activity and accelerate reproduction. Meanwhile, RS, DRS, LRS, and FRS amendments affected the soil microbial community differently, thus biological factors play a non-negligible role in the changes in soil bioavailability of Hg and MeHg after aging straw amendment.

3.4.2. Microbial methylation analysis

Gene copy numbers of hgcA were quantified using qPCR relative to 16 S rRNA. As shown in Fig. S3a, the abundances of the hgcA gene decreased from 4 h to 20 days for RS, DRS, and LRS treatments; However, there was no significant change in FRS, and little variation was observed among the four treatment groups. Furthermore, there was a significant negative correlation between bioavailable MeHg and hgcA abundance (Fig. S3b). This result is not surprising because the more abundance of the hgcA gene does not imply a higher MeHg concentration (Gfeller et al., 2022). Although hgcA gene represents the methylation capacity of the microbial community (Christensen et al., 2019), other factors affect MeHg levels in paddy soil such as demethylation (including microbial demethylation) and soil indicators.

Community barplot analysis was employed to compare the differences between the RS, DRS, LRS, and FRS treatment groups. At the Family level (Fig. 4a), there was a significant difference in the abundance of Clostridiaceae on day 20 ($p < 0.01$). The abundance in DRS, LRS, and FRS treatments decreased by 50.2%, 65.3%, and 53.7%, respectively, compared to RS treatment. The abundance of Clostridiaceae, a typically Hg-methylated bacterium (Zhou et al., 2022), was significantly positively correlated with the concentration of DGT-MeHg ($r = 0.85$, $p < 0.01$) (Fig. S4a). Besides, the most noteworthy change was in the Geobacteraceae, whose growth rate was the highest among all species. The RS, DRS, LRS, and FRS treatments increased from nearly 0 percent at 4 h to 4.1%, 7.0%, 10.2%, and 8.6% respectively at day 20. Previous partial studies have demonstrated that Geobacteraceae is a typical iron-reducing bacterium, that only exists under strict anaerobic conditions, and many strains have been shown to possess a strong Hg methylation capacity (Wang et al., 2022; Bravo et al., 2018). Although the trend of increase was most significant ($p < 0.01$), the abundance of Geobacteraceae was much lower than that of Clostridiaceae, and changes in bioavailability of MeHg among the four treatment groups were mainly due to differences in the abundance of Clostridiaceae rather than Geobacteraceae.

Firmicutes and Actinobacteriota are key bacteria for the methylation and demethylation of Hg, respectively (Capo et al., 2023; Zhou et al., 2020). At the Phylum level (Fig. 4b), the percentage of community abundance for Firmicutes in RS, DRS, LRS, and FRS treatments was

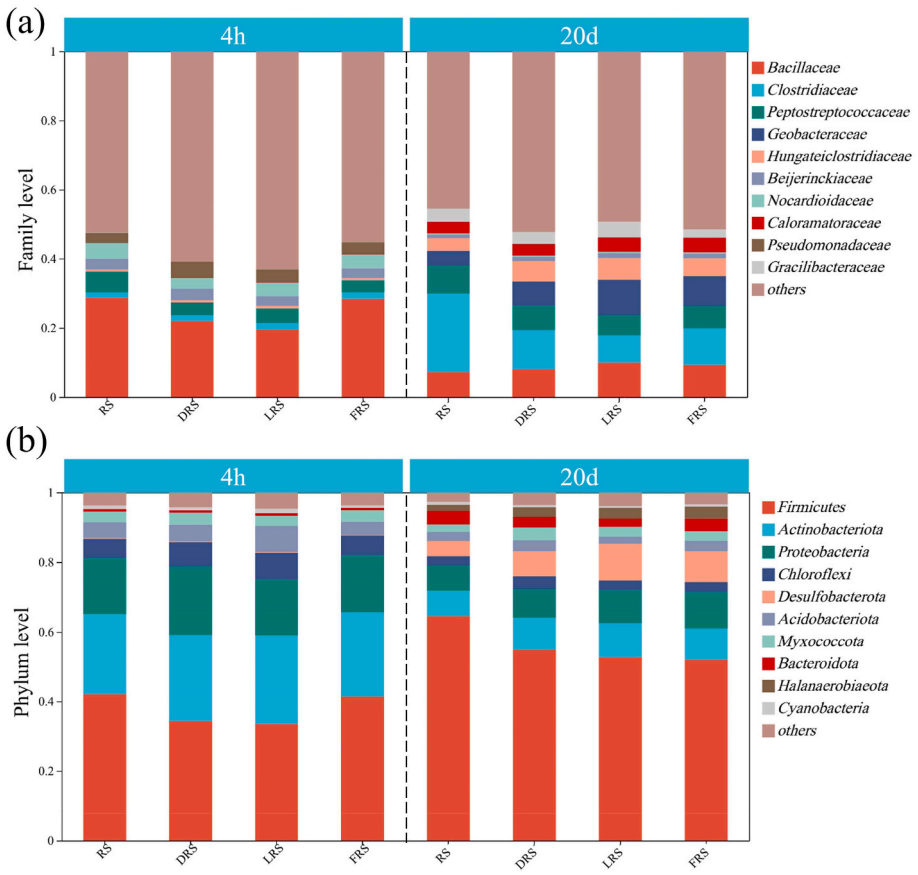


Fig. 4. Percent of community abundance on Family (a) and Phylum (b) levels in paddy soil.

0.645, 0.548, 0.526, and 0.519, respectively on day 20, while that for Actinobacteriota was 0.072, 0.092, 0.098, 0.090, respectively. The abundance of Firmicutes was positively correlated with the DGT-MeHg in soil ($r = 0.81$, $p < 0.05$) (Fig. S4b), while Actinobacteriota showed a negative correlation ($r = -0.72$, $p < 0.05$) (Fig. S4c). Therefore, RS, DRS, LRS, and FRS amendment can alter soil MeHg concentrations by changing the abundance of methylated microorganism Firmicutes and demethylated microorganism Actinobacteriota.

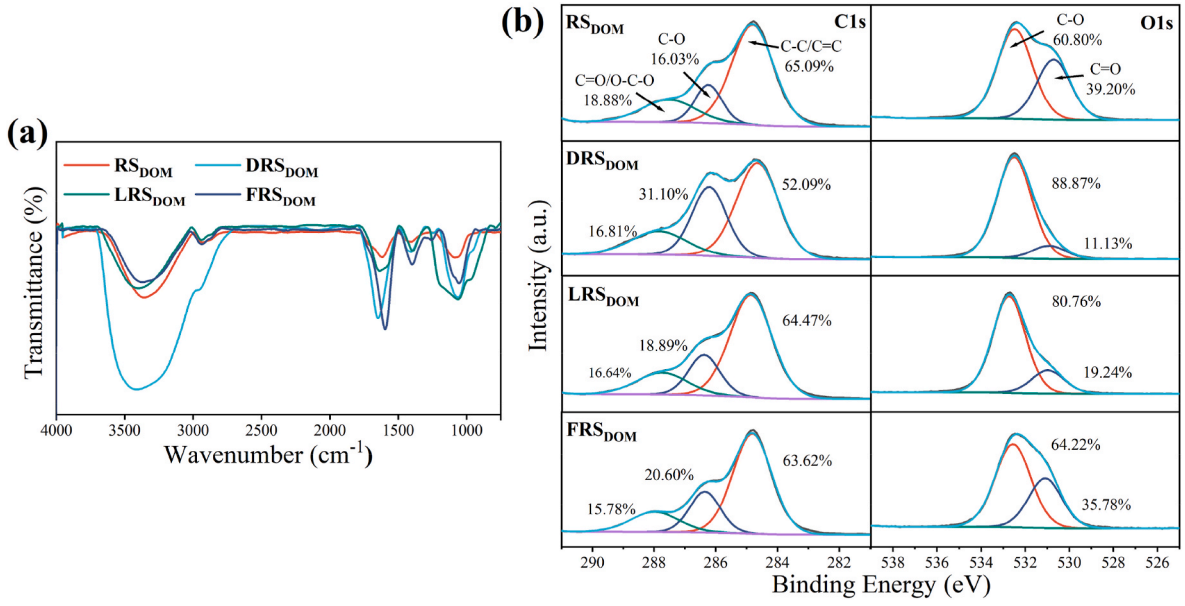


Fig. 5. Fourier transform infrared spectra (FTIR) (a) and X-ray photoelectron microscopy (XPS) (b) of fresh, dry-wet aging, photochemical aging, and freeze-thaw aging straw-DOM.

3.5. Effect of straw-DOM on the bioavailability of Hg

3.5.1. Changes in the properties of aging straw-DOM

Compared to RS_{DOM}, the stronger signals observed in DRS_{DOM}, LRS_{DOM}, and FRS_{DOM} at 1637 cm^{-1} suggest that straw-DOM is predominantly composed of small molecules (Ma et al., 2015) (Fig. 5a). Similarly, the O: C atomic ratios of RS_{DOM}, DRS_{DOM}, LRS_{DOM}, and FRS_{DOM} were 0.63, 0.64, 0.99, and 0.43, respectively, indicating that light irradiation caused the straw leaching solution to become more oxidized (Table S6). According to Fig. 5b, the content of C═O in RS_{DOM}, DRS_{DOM}, LRS_{DOM}, and FRS_{DOM} was 39.20%, 11.13%, 19.24%, and 35.78%, respectively, for the O1s spectrum. Additional analysis information about XPS is available in Text. S6. In summary, significant changes have occurred in the surface structure, functional group composition, and content of straw and straw-DOM after being treated with dry-wet, freeze-thaw, and photochemical aging.

3.5.2. Increased inorganic Hg (IHg) uptake by seedlings

To avoid interference from straw adsorption, soil properties, and Hg-methylated microbes, sand culture experiments were conducted to explore the net effect of aging straw-DOM on the bioavailability of Hg in paddy soils. Since soil bioavailable Hg measured by DGT has been widely used to predict the ability of IHg uptake by plants, thus the results of IHg uptake by rice seedlings in sand culture experiments will be used to compare with the DGT-Hg in incubation experiments. The positive effects of DRS_{DOM}, LRS_{DOM}, and FRS_{DOM} on IHg uptake by rice seedlings may be due to two reasons: (1) the effect on the amount of IHg uptake by rice seedlings through affecting their tip number and root length; and (2) variations of the optical properties and functional groups of the DRS_{DOM}, LRS_{DOM}, and FRS_{DOM} that affect Hg concentrations in solution and changing the amount absorbed by rice seedlings.

As shown in Fig. 6a, the ΔHg concentrations in DRS_{DOM}, LRS_{DOM}, and FRS_{DOM} treatments were 118.2%, 35.6%, and 217.1% higher than those in RS_{DOM} treatment on day 10, and 97.1%, 28.1%, 198.0% higher than those in RS_{DOM} treatment on the day 20, respectively. This result indicates that DRS_{DOM}, LRS_{DOM}, and FRS_{DOM} can markedly enhance IHg uptake by rice seedlings, with the most significant increase of nearly two-fold in FRS_{DOM}. Previous research has confirmed that the addition of straw-DOM can enhance IHg uptake by seedlings (Wang et al., 2020), and this phenomenon is more pronounced in FRS_{DOM}, especially when aging straw-DOM is added. However, the results showed that the tip number and root length of rice seedlings cultured with DRS_{DOM}, LRS_{DOM}, and FRS_{DOM} did not differ significantly compared with the RS_{DOM} (Figs. S5a and b), thus the tip number and root length were not factors affecting the IHg uptake by rice seedlings in this experiment. After excluding the role of root length and root tip, the focus will be placed on the effect of differences in properties of DRS_{DOM}, LRS_{DOM}, and FRS_{DOM} on IHg uptake by rice seedlings in the following.

According to the results of EEM, as per Simpson's Rule (Zhou et al., 2013), dividing the fluorescent zone into five zones to discern the composition of the DOM. As shown in Fig. S6, the FRS_{DOM} exhibits significantly stronger fluorescence intensity in the EEM spectral regions I and II than others, indicating that it contains more aromatic proteins (Tan et al., 2019), reflecting greater bioavailability (Inamdar et al., 2012). Additionally, DRS_{DOM}, LRS_{DOM}, and FRS_{DOM} show stronger fluorescence intensity than RS_{DOM} in region IV (soluble microbial byproduct-like materials). Correlations between Hg levels in rice seedlings and DOM properties were assessed, and the results are shown in Fig. 6b. The value of E_2/E_3 ($A_{250\ nm}/A_{365\ nm}$) is significantly positively correlated with Hg on days 10 and 20 ($p < 0.05$). According to Table S3, the E_2/E_3 value of DOM for RS, DRS, LRS, and FRS are 4.284, 5.627, 4.567, and 6.209, respectively. A higher value of E_2/E_3 represents DOM with a lower molecular weight, which represents a better ability to form complexes with Hg and increase migration from solution to rice seedlings (Leresche et al., 2022; Su et al., 2021). This result is consistent with FTIR spectra. Compared to RS_{DOM}, the DRS_{DOM}, LRS_{DOM}, and FRS_{DOM} had more intense signals at 1637 cm^{-1}, composed of primary carbohydrates with small molecules, thus enhancing the migration of Hg. Furthermore, the E_2/E_3 ratio was positively correlated with E_3/E_4, and compared to RS_{DOM}, the E_3/E_4 ratio of DRS_{DOM} and FRS_{DOM} increased by 70.4% and 23.2%, respectively. These results indicate that the content of fulvic acids increased in the DRS_{DOM} and FRS_{DOM} after dry-wet aging and freeze-thaw aging, which has been confirmed to represent better IHg uptake by seedlings (Artinger et al., 2000; Xu et al., 2021). Furthermore, the XPS results of RS_{DOM}, DRS_{DOM}, LRS_{DOM}, and FRS_{DOM} showed that the O: C atomic ratios (represent O-containing functional groups) were 0.63, 0.64, 0.99, and 0.43, respectively, and C═O functional group content was 39.20%, 11.13%, 19.24%, and 35.78%, respectively. Compared to RS_{DOM}, DRS_{DOM}, LRS_{DOM}, and FRS_{DOM} have relatively low C═O content, indicating their inferior ability to combine with Hg and more IHg uptake by rice seedlings (Ma et al., 2015). However, the results were opposite to what was expected, as DRS_{DOM} and LRS_{DOM} with lower C═O =content should be shown a high ΔHg value compared to FRS_{DOM}. This may be caused by the presence of more O-containing functional groups in DRS_{DOM} (0.64) and LRS_{DOM} (0.99) compared to FRS_{DOM}, indicating that other functional groups may play a role in Hg fixation. Therefore, it is reasonable to assume that the decrease in C═O functional group content is an important factor

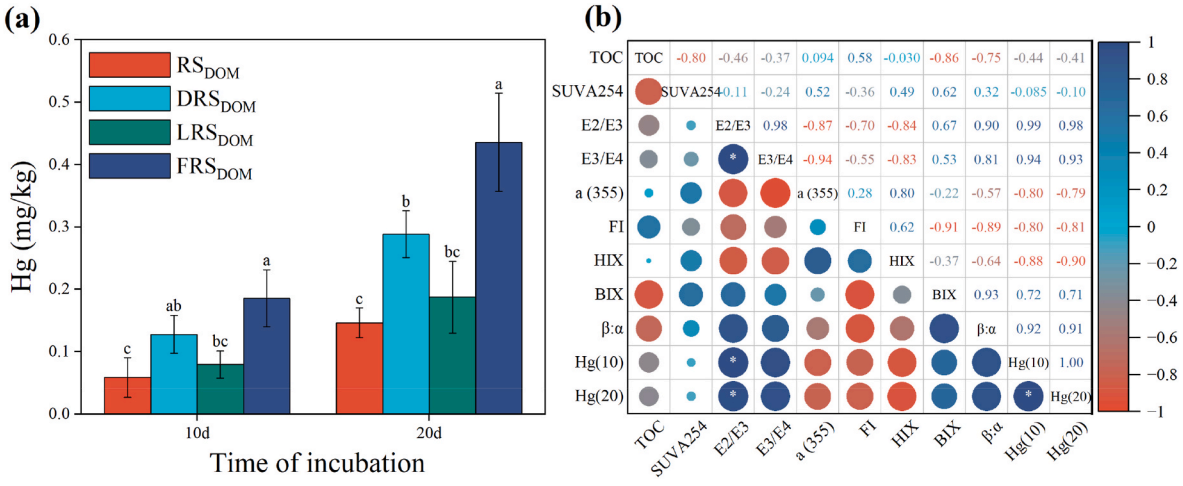


Fig. 6. The ΔHg (elevated amount relative to the pure water incubation group) absorbed by seedlings (a); Correlations between total Hg in seedlings and DOM properties from different straws (b). * and ** represent significant correlation at $p < 0.05$ and $p < 0.01$ level, respectively. Data are given as means ± SD (n = 3). Different letters indicate significant differences among treatments within the same day ($p < 0.05$).

promoting rice seedling uptake of IHg by aging straw-DOM.

4. Conclusions and environmental implications

Our present results indicate that the bioavailability of Hg and MeHg in paddy soil is reduced after aging straw (DRS, LRS, and FRS) amendment, while aging straw-DOM (RS_{DOM}, DRS_{DOM}, LRS_{DOM}, and FRS_{DOM}) has an opposite effect. Firstly, in terms of soil properties, the DRS, LRS, and FRS decompose in paddy soil, enhancing soil pH and reducing dissolved Fe and Mn in the overlying water, ultimately inhibiting Hg methylation. Secondly, in terms of biological factors, amendments with DRS, LRS, and FRS reduce the MeHg bioavailable in paddy soil by suppressing the activity and reducing the abundance of methylating microorganisms (Clostridiaceae and Firmicutes) and promoting the activity and increasing the abundance of demethylating microorganisms (Actinobacteriota). Moreover, our research found that amending with LRS and FRS reduces the Hg bioavailable in paddy soil compared to RS amendment, which decreases with decreasing SO_4^{2-} content in paddy soil. However, the DRS amendment showed an increased trend in the bioavailability of Hg compared to RS amendment. Besides, the DRS_{DOM}, LRS_{DOM}, and FRS_{DOM} were found to enhance the absorption of Hg in rice seedlings, suggesting that aging straw-DOM could promote the bioavailability of Hg. We characterized DOM and found that dry-wet aging, photochemical aging, and freeze-thaw aging decreased the molecular weight and C═O functional group content of DOM, thus promoting the free nature of Hg and its migration to rice seedlings.

The main aging processes that straw undergoes when it is returned to paddy soil vary in different climatic regions of China. The straw amendment is beneficial for reducing the bioavailability of Hg and MeHg in paddy soil in areas with sufficient light and significant temperature fluctuations. On the contrary, we should minimize straw amendment in regions with high rainfall to prevent an increase in the bioavailability of Hg in paddy soil. Furthermore, straw-DOM has the opposite impact on the bioavailability of Hg in paddy soil compared to straw, which reminds us to be aware of the potential role of straw-DOM in increasing the exposure risk of Hg in paddy soil. The results of this study provide important references and guidance for further improving straw management practices.

Credit authorship contribution statement

Yubo He: Investigation, Formal analysis, Resources, Data curation, Methodology, Writing - Original Draft, Xu Yang: Methodology, Formal analysis, Resources, Zihao Li: Software, Data curation, Tantan Wang: Methodology, Resources, Chi Ma: Formal analysis, Xin Wen: Software, Wenhao Chen: Software, Chang Zhang: Funding acquisition, Writing - Review & Editing.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.chemosphere.2023.139711.

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